EMCH 201: Introduction to Application of Numerical Methods

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Assignment 2

Assignment due date: **11:59 PM, Monday September 20th , 2021**
Point Possible: 100 (30+30+40)

Submission requirements:

a. Detailed problem-solving process needs to be provided for full points
b. Upload readable electronic copy to blackboard in a single file
c. Late submission will not be graded

1. Determine the root of $f(x) = 3x^3 - 2x^2 + 6x - 200$ using the bisection method. The result has at least 1 significant figure. Employ initial guesses of x_l = 3.0 and x_u = 5.0. (HINT: determine the pre-specified percent tolerance/error based on the required significant figures) (30 pts)

2. For a function $f(x) = e^{-x^2} - x$, if the initial guess (x_i) is 0.6 then what will be the value of x_{i+1} using Newton-Raphson method after two iterations? Calculate the approximate percent relative error ε_a for each iteration. (30 pts)

3. Use the Newton-Raphson method to determine the drag coefficient c for a parachutist of mass = 68.1 kg in order to reach a velocity of 40 m/s when time t = 10s. Using an initial guess of c = 15 kg/s and obtaining approximate percent relative error less than 0.1%. The relationship between the velocity and the time is given by $v = \frac{gm}{c}\left(1 - e^{-\left(\frac{c}{m}\right)t}\right)$. The other parameters are given below (40 pts).

Variables	Annotation	Value
m	Parachutist mass	68.1 kg
t	Time	10 s
g	Gravity	9.81 m/s^2
v	Velocity	40 m/s